Harnessing the Power of Ensemble Algorithms for Diabetes Prediction: A Comparative Analysis

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***Abstract*—Effective diagnosis of diabetes is crucial for managing the disease and preventing complications. This study explores the use of machine learning for diabetes prediction, focusing on the impact of data preprocessing on model performance. Using four classifiers, namely logistic regression, naive Bayes, random forest, and gradient boosting, diabetes cases are diagnosed after extensive data preprocessing, including cleaning and feature engineering. Results show that the ensemble methods, random forest, and gradient boosting achieve high accuracy and robustness in diabetes prediction, especially with preprocessing. These findings underscore the importance of model selection and preprocessing in healthcare machine learning.**

***Index Terms*—Diabetes Prediction, HbA1c, Smoking History, Logistic Regression, Naive Bayes, Random Forest, XGBoost**

I. Introduction

Diabetes is a chronic illness characterized by elevated blood glucose levels, which is caused by the body's inability to effectively use insulin or the pancreas' inability to produce enough insulin [1]. This leads to a disturbance in the metabolism of fats and proteins. There are three major types of diabetes: Type 1, where the pancreas does not produce insulin; Type 2, where the body cells are resistant to the action of insulin that is being produced and over time the production of insulin progressively decreases; and gestational diabetes, which occurs during pregnancy and can cause some complications during the pregnancy, and at birth and increases the risk of Type 2 diabetes in the mother and obesity in the offspring. In addition to these types of diabetes, there are two other categories of glucose intolerance - Impaired Fasting Glucose (IFG) and Impaired Fasting Glycemia (IGT) - which are intermediate conditions between normal and diabetic blood glucose levels. However, the transition to diabetes is not inevitable. People with IFG and IGT are at an increased risk of cardiovascular disease than people with normal blood glucose values. This paper uses different machine learning models to compare their approximate accuracy with different parameters that increase the likelihood of diabetes [2][3].

The literature review explores the intersection of diabetes research, machine learning, and artificial intelligence through critical studies and reports. Forouhi and Wareham (2010) provide foundational insights into diabetes epidemiology, emphasizing its prevalence, risk factors, and public health implications [3]. Nikam (2015) conducts a comparative study of machine learning techniques in diabetes research, emphasizing the importance of accurate predictive models for diagnosis and prognosis [4]. Chen and Guestrin (2016) introduce XGBoost, a scalable tree-boosting system widely used in healthcare, including diabetes research, to build robust predictive models [5]. Roglic (2016) summarizes the WHO Global Report on diabetes, offering crucial findings and recommendations for addressing the global epidemic [2]. Sherwani et al. (2016) underscore the significance of the Glycated Hemoglobin (HbA1c) test in diabetes management, highlighting its role in monitoring blood glucose levels over time for better treatment outcomes [6]. Boateng et al. (2019) review the logistic regression model's applications in medical research, particularly in predicting patient outcomes and identifying risk factors in diabetes studies [7]. Kumar et al. (2023) conducted a systematic literature review on artificial intelligence in disease diagnosis, highlighting its potential to enhance diagnostic processes in diabetes and other medical conditions. This review emphasizes the multidisciplinary nature of diabetes research, integrating insights from epidemiology, clinical diagnostics, machine learning, and artificial intelligence to advance the understanding of the disease and improve patient care outcomes globally [1].

II. Methodology

The methodology employed in the study is represented in Figure 1

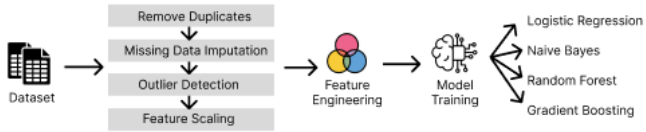

Fig. 1: Proposed Workflow

A. Data Acquisition

The current study utilizes the Diabetes_ Prediction_Dataset available on the public platform. This dataset comprised 100,000 rows and nine columns, which included various features associated with diabetes diagnoses like gender, age, hypertension, heart disease history, smoking history, bmi (Body Mass Index), HbA1c level, blood glucose level, and diabetes as demonstrated in Table I.

TABLE I: Raw Data

Gender	Age	Hypertension	Heart Disease	Smoking History	BMI	HbA1c Level	Blood Glucose Level	Diabetes
Female	80	0	1	Never	25.19	6.60	140	0
Female	54	0	0	No Info	27.32	6.60	80	0
Male	28	0	0	Never	27.32	5.70	158	0
Female	36	0	0	Current	23.45	5.00	155	0
Male	76	1	1	Current	20.14	4.80	155	0

B. Data Pre-processing

1) Data Cleaning: Before analysis, rigorous data cleaning procedures were performed to enhance the quality of the dataset. This involved the removal of null and duplicate rows to ensure data integrity. This resulted in a data set of 96,128 rows and nine columns.

2) Data Binning: To better capture the underlying patterns in blood glucose levels, binning was performed, a technique that divides the continuous variable into discrete intervals. This process facilitates the identification of significant thresholds and enhances model interpreting ability.

3) Data Transformation: Categorical variables were encoded using one-hot encoding to convert them into numerical format, facilitating model training. The cleaned and transformed data is illustrated in Table II.

4) Handling Outliers: Clipping was applied to the BMI feature to limit extreme values. By setting upper and lower bounds, the influence of outliers was mitigated and ensured the stability of the models.

5) Feature Engineering: Feature engineering plays a crucial role in enhancing the predictive power of machine learning models. The following techniques were employed to facilitate feature engineering.

- **Correlation Analysis**: Correlation plots were developed to identify relationships between different features and the target variable as presented in Figure 2.

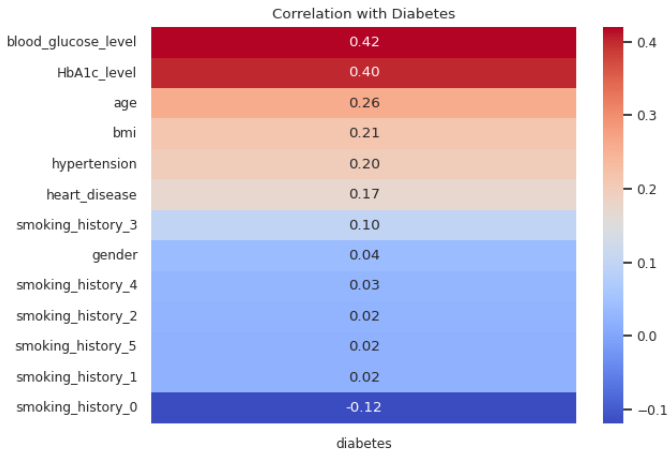

Fig. 2: Correlation of Pre-processed Data

- **Feature Reduction**: To streamline the dataset and improve model efficiency, the total number of features was reduced by removing irrelevant features and merging similar ones. The number of classes in the smoking feature was reduced by merging categories based on their relationship with the target feature. This step aimed to simplify the feature while preserving its predictive capacity.

6) Data Balancing: Addressing class imbalance is vital for the accurate prediction of minority classes. To counter class imbalance within the acquired dataset, the third gender category was removed from the dataset. This decision was made to ensure balanced representation across gender categories and to prevent potential biases in model training. Also, the Synthetic Minority Over-sampling Technique (SMOTE) was employed. SMOTE generates synthetic instances of the minority class by interpolating between existing minority class samples, thereby balancing class distribution. This technique helps prevent the model from being biased towards the majority class, thus enhancing the model's ability to generalize to unseen data.

C. Impact of Data Pre-processing

Significant improvements were observed in feature correlation after applying various preprocessing techniques as displayed in Figure 3. By comparing Figure 2 and 3, it can be

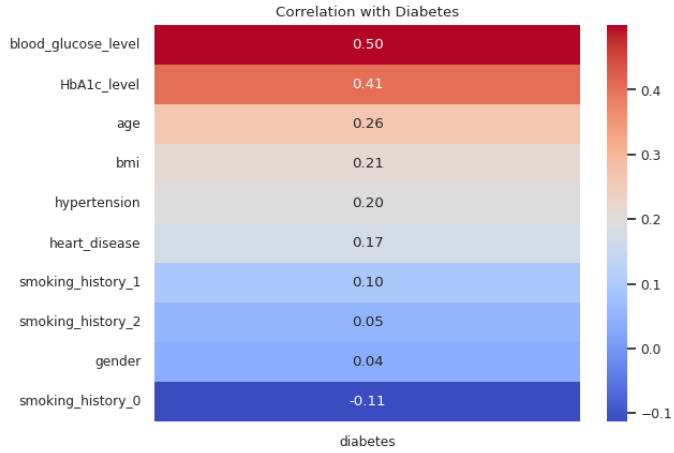

Fig. 3: Correlation Enhancement

observed that the correlation of blood glucose level increased from 0.42 to 0.50, correlation of HbA1c level improved from 0.40 to 0.41, correlation of smoking history was considerably

TABLE II: Preprocessed Data

Gender	Age	Hypertension	Heart Disease	BMI	HbA1c Level	Blood Glucose Level	Diabetes	Smoking History 0	Smoking History 1	Smoking History 2
0	80	0	1	25.19	6.60	0	0	0	0	1
0	54	0	0	27.32	6.60	0	0	1	0	0
1	28	0	0	27.32	5.70	0	0	0	0	1
0	36	0	0	23.45	5.00	0	0	0	0	1
1	76	1	1	20.14	4.80	0	0	0	0	1

enhanced, indicating a stronger relationship with the target variable.

D. Model Training

The dataset was divided into train and test sets maintaining a 8:2 ratio. Hyperparameter tuning was then performed to optimize the performance of machine learning models. Grid search, a systematic approach to hyperparameter tuning was utilized, that exhaustively searches through a predefined hyperparameter grid to identify the optimal combination of hyperparameters. This technique helps fine-tune model parameters and improve model performance.

The machine learning models employed for the training are explained in the subsequent subsection.

1) Gaussian Naive Bayes: The algorithm is based on the principle of Bayes theorem from probability theory and functions. Every parameter of the data set is independent and it follows a Gaussian distribution. The algorithm then calculates the variance means for each feature (y) of the data set and then calculates the probability of each feature (x) [4].

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} \tag{1}$$

$$P(x_i|y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\nu)^2}{2\sigma^2}} \tag{2}$$

2) Logistic Regression: It is used for binary classification, also known as the sigmoid function, which takes input as the independent variable and gives output the probability in either 0 or 1 [7][8]. The sigmoid function is given in Equation 3.

$$\sigma(z) = \frac{1}{1 + \exp^{-z}} \tag{3}$$

$$z = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n \tag{4}$$

3) Random Forest: The utilization of the random forest algorithm is ideal for resolving large-scale problems by combining predictions from randomized decision trees. In this review, the study delves into the latest advancements in the algorithm, including parameter selection, resampling, and variable importance. The algorithm creates M-randomized trees, halting growth when the cell count falls below the designated node size. Each regression tree predicts the average of Y_i values, corresponding to X_i falling within the x cell.

$$\hat{y}_i = \frac{1}{N} \sum_{j=1}^{N} T_j(x_i) \tag{5}$$

where:
$\hat{y}_i$ is the predicted value for sample i.
N is the number of trees in the random forest.
T_j represents the j-th decision tree in the ensemble.
x_i is the feature vector for sample i.
Each decision tree T_j makes an individual prediction for the input feature vector x_i. The final prediction $\hat{y}_i$ is obtained by averaging the predictions of all decision trees in the ensemble.

The hyperparameter tuning values for the random forest are illustrated in Table III.

TABLE III: Hyperparameter Tuning Values

Hyperparameters	Values
Learning Rate	[0.01, 0.1]
Max Depth	[3, 7]
Max Feature	[0.1, 1]
Minimum Sample Split	[2, 10]
n-estimator	[50, 200]

4) Gradient Boosting (XGBoost): The gradient boosting algorithm builds an ensemble of weak learners, typically decision trees, by sequentially adding models that correct the errors of the previous ones. Unlike random forests, which build trees independently and aggregate their predictions, gradient boosting focuses on reducing the residual errors by training each new tree on the gradients (i.e., the errors) of the loss function. This iterative process continues until the model reaches a predefined number of trees or the residual errors cannot be reduced further. The algorithm's performance can be fine-tuned through parameters such as the learning rate, tree depth, and the number of boosting rounds. Gradient boosting offers high predictive accuracy, especially in cases where the relationships between features and the target variable are complex [5][8].

$$\hat{y}_i = \sum_{k=1}^{K} \gamma_k f_k(x_i), \quad f_k \in \mathcal{F} \tag{6}$$

where:
$\hat{y}_i$ is the predicted value for sample i.
K is the number of trees.
f_k represents the k-th tree in the ensemble.
$\mathcal{F}$ is the space of possible trees.

The hyperparameter tuning values for the gradient boosting are mentioned in Table III.

III. Results and Discussion

The performance of machine learning models was compared on the complete feature set and reduced feature set, both before and after hyperparameter tuning. The model performance was evaluated using accuracy, Area Under the Curve (AUC) score, F1-score, recall, and Receiver Operating Characteristic (ROC) curve. These metrics helped assess the discriminative power of the models and visualize their performance across different thresholds.

A. Results without Data Pre-processing

Table IV presents the results obtained without performing any preprocessing steps on the dataset for diabetes prediction. Notably, both random forest and gradient boosting classifiers maintain high performance, with an accuracy score of 97.2% and an AUC score of 0.837 and 0.842, respectively. These models outperform naive Bayes and logistic regression in all performance metrics.

TABLE IV: Results without Data Pre-processing

Model	Accuracy	AUC	Recall	F1 Score
Logistic Regression	0.957	0.794	0.599	0.702
Naive Bayes	0.904	0.790	0.653	0.537
RandomForest	0.972	0.837	0.674	0.805
Gradient Boosting	0.972	0.842	0.685	0.809

B. Results with Data Preprocessing

Table V and Figure 4 display the performance of four different models for diabetes prediction after data preprocessing. It is observed from Table V that both random forest and gradient boosting classifiers achieve a high accuracy score of 96.9%, along with comparable AUC scores of 0.826 and 0.828, respectively. These models also exhibit similar recall and F1-score values, indicating their effectiveness in correctly identifying instances of diabetes. It can be concluded that the preprocessing steps contributed to enhancing the predictive performance of these ensemble methods by reducing noise, handling missing values, and extracting relevant features.

TABLE V: Results with Data Pre-processing

Model	Accuracy	AUC	Recall	F1 Score
Naive Bayes	0.890	0.795	0.680	0.521
Logistic Regression	0.958	0.800	0.608	0.719
RandomForest	0.969	0.826	0.653	0.790
Gradient Boosting	0.969	0.828	0.657	0.790

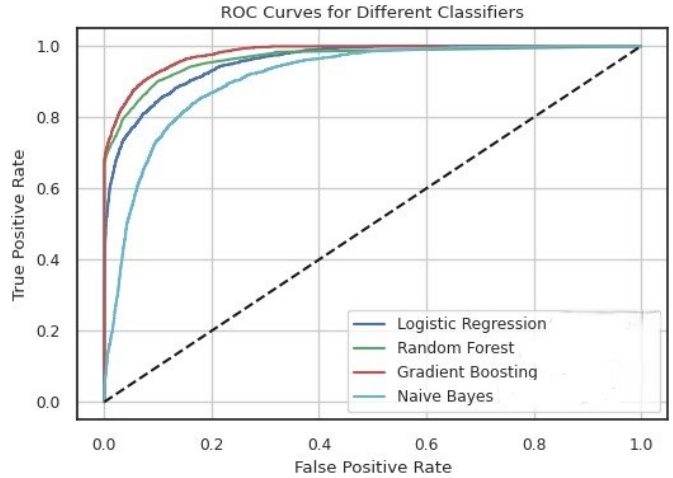

Fig. 4: ROC Curves

In contrast, the naive Bayes classifier shows lower performance with an accuracy of 89.0% and lower AUC, recall, and F1 scores. Despite its simplicity and assumption of feature independence, naive Bayes struggles to capture the underlying patterns in the data effectively. The preprocessing steps have a limited impact on naive Bayes due to its inherent assumptions, highlighting the importance of choosing appropriate models based on data characteristics [4].

The logistic regression model achieves an accuracy of 95.8% and an AUC score of 0.800, which is lower than the ensemble methods. Table V demonstrates that the recall and F1 score values are also lower than the ensemble models, suggesting its unsuitability for this classification task.

C. Feature-wise Analysis for Diabetes Diagnosis

The following statistical representations show the variation of diabetic patients with various features used in the study.

1) Variation with Gender: Figure 5 shows the diabetes diagnosis variation with gender. It is evident from Figure 5 that there are more female patients than male patients with a ratio of 20:19 for diabetes with gender distribution.

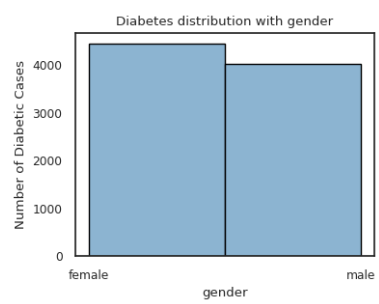

Fig. 5: Gender Distribution

2) Variation with Age: Figure 6 illustrates the variation in diabetes diagnoses across different age groups. The majority

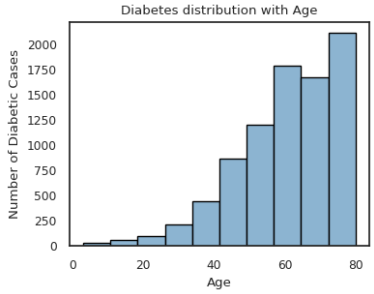

Fig. 6: Age Distribution

number of diagnosed cases of diabetes was found among the age groups of 60-80 years old. It is observed that patients' age curve shows an increasing trend, showing that patients at younger ages are less likely to get diabetes.

3) Variation with Hypertension: Figure 7 depicts the variation in diabetes diagnoses among patients with hypertension. Around 2,000 patients with hypertension as a leading cause

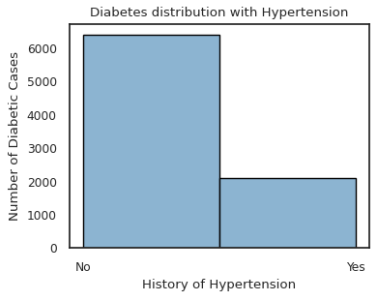

Fig. 7: Hypertension Distribution

were diagnosed with diabetes, whereas 6,000 did not have diabetes.

4) Variation with Heart Disease History: Figure 8 illustrates the variation in diabetes diagnoses among patients with a previous history of heart diseases. Out of the patients diagnosed

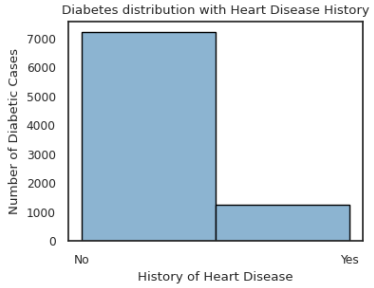

Fig. 8: Heart Disease History Distribution

with some heart disease, around 1,500 were also diagnosed with diabetes, while roughly 7,000 showed no signs of diabetes.

5) Variation with Smoking History: Figure 9 shows the variation of diabetic patients with smoking as a habit. Figure 9

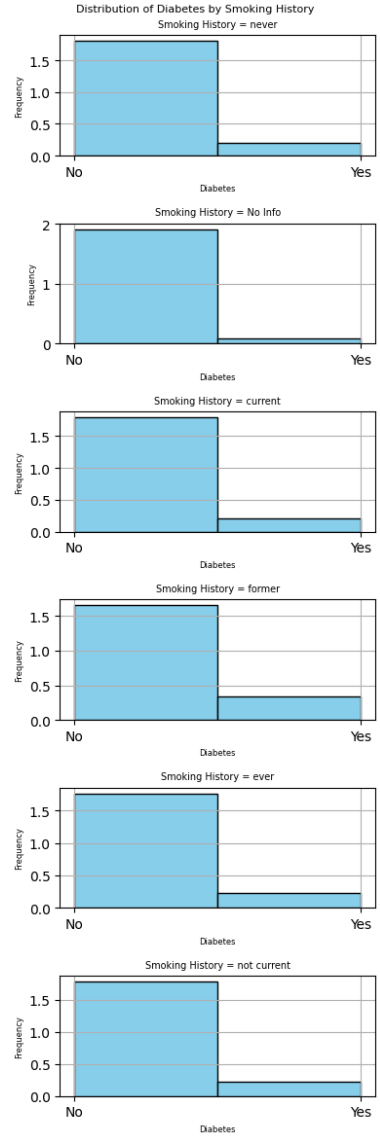

Fig. 9: Smoking History

reveals that 1 out of 6 patients with no history of smoking has been diagnosed with diabetes. 1 out of 6 patients who have had smoking as a habit has been diagnosed with diabetes. Nearly 1 out of 10 patients have been diagnosed with diabetes.

D. Variation with BMI

Figure 10 shows the variation in diabetes diagnoses with the effect of BMI. The majority cases of diagnosed cases of diabetes were found among the patients with BMI varying from 30-40 which shows a bell curve distribution.

E. Variation with HbA1c Level

Figure 11 illustrates how diabetes diagnoses vary with changes in HbA1c levels. The distribution indicates that the highest number of diabetic patients falls within the HbA1c range of 6.0-7.0, with levels ranging from 250 to 1250.

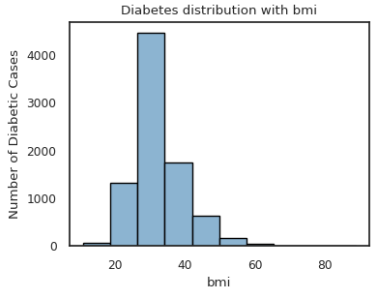

Fig. 10: BMI Distribution

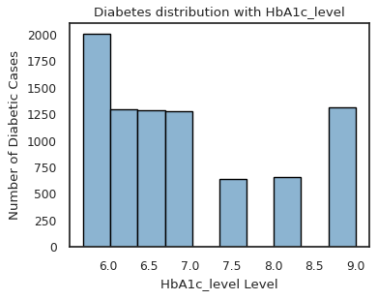

Fig. 11: HbA1c Level Distribution

F. Variation with Blood Glucose Level

Figure 12 shows the variation in diabetes diagnoses associated with different blood glucose levels. The majority cases of

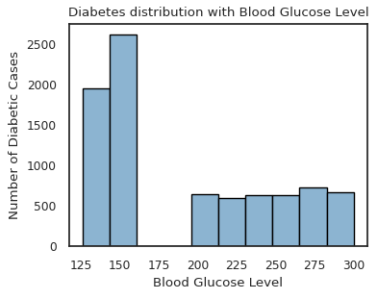

Fig. 12: Blood Glucose Distribution

diagnosed cases of diabetes were found among patients with a blood sugar level of 150. In contrast, for a level of 200-300, the number of patients is relatively constant at the range of 500.

IV. CONCLUSION

This study investigated the application of machine learning models for the prediction of diabetes, focusing on the effect of preprocessing for enhanced model performance. The results of the experiments demonstrate the effectiveness of ensemble methods in accurately predicting diabetes. These models consistently achieved high accuracy and ROC-AUC scores, indicating their robustness and reliability in classifying diabetes cases. The preprocessing steps played a crucial role in improving the performance of these models by reducing noise, handling missing values, and extracting relevant features. The findings contribute to the growing body of research on the application of machine learning in healthcare and underscore the potential for improving patient care through data-driven approaches.

Further research can explore alternative modeling approaches and the integration of additional features to enhance the performance of diabetes prediction. Additionally, evaluating the generalizability and scalability of these models on larger and more diverse datasets will be crucial for their real-world deployment and clinical application.

REFERENCES

[1] Y. Kumar, A. Koul, R. Singla, and M. F. Ijaz, "Artificial intelligence in disease diagnosis: a systematic literature review, synthesizing framework and future research agenda," *Journal of ambient intelligence and humanized computing*, vol. 14, no. 7, pp. 8459–8486, 2023.

[2] G. Roglic, "Who global report on diabetes: A summary," *International Journal of Noncommunicable Diseases*, vol. 1, no. 1, pp. 3–8, 2016.

[3] N. G. Forouhi and N. J. Wareham, "Epidemiology of diabetes," *Medicine*, vol. 38, no. 11, pp. 602–606, 2010.

[4] S. S. Nikam, "A comparative study of classification techniques in data mining algorithms," *Oriental Journal of Computer Science and Technology*, vol. 8, no. 1, pp. 13–19, 2015.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '16. New York, NY, USA: ACM, 2016, pp. 785–794. [Online]. Available: http://doi.acm.org/10.1145/2939672.2939785

[6] S. I. Sherwani, H. A. Khan, A. Ekhzaimy, A. Masood, and M. K. Sakharkar, "Significance of hba1c test in diagnosis and prognosis of diabetic patients," *Biomarker insights*, vol. 11, pp. BMI–S38 440, 2016.

[7] E. Y. Boateng, D. A. Abaye *et al.*, "A review of the logistic regression model with emphasis on medical research," *Journal of data analysis and information processing*, vol. 7, no. 04, p. 190, 2019.

[8] A. Géron, *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*. " O'Reilly Media, Inc.", 2022.